

How GenAI Endangers Indigenous Artworks: A Rights and Justice Lens

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Abstract

Indigenous Peoples¹ have historically faced cultural appropriation of their art, but the issue becomes even more pertinent in the age of AI. The ongoing *Andersen v. Stability AI* lawsuit will be a landmark case in copyright litigation, specifically examining how AI-image tools learn and what that means for the rights of creative artists. In this paper, we focus an interdisciplinary lens on how the emergence of AI-image tools places disproportionate burdens on Indigenous Peoples without adequately sharing the benefits of AI technology with them. We use two descriptive case studies to understand why the current IP regime fails to adequately protect artists and present some plausible solutions to navigate upcoming challenges.

1 The term 'Indigenous Peoples' with capitalized initials is based on the capitalization used in Gregory Younging's 2018 book. Moreover, the term is capitalized in this manner following the United Nations Editorial Manual, updated in accordance with General Assembly resolution 77/203. The author recognizes that there are multiple different accepted capitalizations of the term.

Keywords: Artificial Intelligence, Indigenous Art, AI-Image Tools, GenAI Art

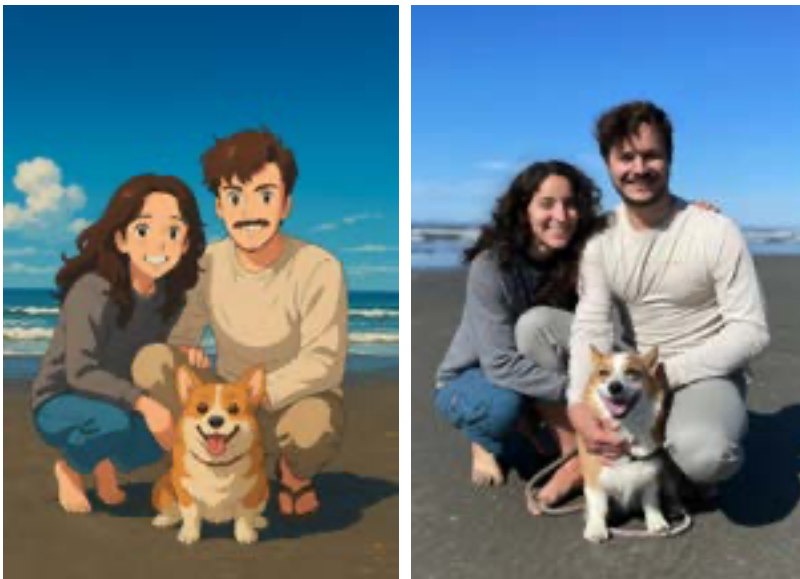
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Introduction

The 2020s will be remembered for normalizing the use of Artificial Intelligence (AI) software in our lives (London School of Economics, 2025). AI chatbots like ChatGPT or Gemini can now answer almost all questions. The impact of AI is also felt in creative spaces, where it is used to write prose and poetry and even to create visual art. Visual art made using Generative AI (hereafter, GenAI) platforms like Midjourney, OpenAI's DALL-E, and Adobe Firefly can alter how we produce and consume visual media. The rise of GenAI platforms for creating visual art has raised concerns about the rise of doctored and outright fake images in digital media. AI software is trained (programmed to learn and mimic patterns and features of data) on images found online, which helps it produce a perfect likeness of the images requested by a user with a 'text-based prompt', i.e., a question, command, or description.

Figure 1

The first image posted online by an X user in March 2025 that started the 'Studio Ghibli art' trend.



Note. Original image on the right, recreated image on the left.

Image sourced from Grant Slatton's post on X.

(<https://x.com/GrantSlatton/status/1904631016356274286?lang=en>)

In April 2025, we witnessed the “Studio Ghibli AI art” trend, in which social media was flooded with Studio Ghibli-inspired images generated by ChatGPT. A user could simply upload their own picture or any original image to ChatGPT and provide a prompt to create a likeness of the Ghibli animation style, after which the AI would recast the image in a style characteristic of the Japanese artist Hayao Miyazaki’s illustrations. Studio Ghibli is an independent animation studio co-founded by Miyazaki that produces animated films, including *My Neighbour Totoro* (1988), *Princess Mononoke* (1997), and *Spirited Away* (2001), all of which are highly acclaimed. This trend brought to light the issue of AI models being trained on artwork that was used without permission from the creators. It is important to note that Studio Ghibli holds the copyright to its films, yet it was not immune to the unauthorized use of its artistic style to train AI models (Spangler, 2025).

Indigenous Peoples are the original inhabitants of a land, who inhabited pre-colonial societies and have a rich cultural history expressed through music, art, and design. Their artworks have a distinct style that is passed down through tradition. The advent of the Internet means more of these artworks and cultural expressions have found their way online through academic research, museum archives, and shared digital spaces. While some of these works are protected, the intellectual property (IP) regime has not granted blanket protections to all such creative works. Even in the absence of AI software, Indigenous imagery was being culturally appropriated by fashion retailers (Baca, 2025), fashion/sports brands (BBC, 2019), and non-Indigenous artists (Yesno, 2025). With the advent of GenAI, which trains itself on all images on the Internet, it becomes pertinent to enquire whether the extant IP protections can rise to the new challenges posed by this evolving technology.

To understand the deeper impacts of unauthorized use of artworks, the paper is laid out as follows:

Part 1 briefly explains how GenAI software create images, Part 2 explores Indigenous Peoples, their rights over data and understanding cultural appropriation, Part 3 are two case studies, Part 4 briefly summarizes learnings from the case studies to understand why current legal frameworks fall short of ensuring fairness for artists, Part 5 centers select ethical concerns related to AI and unauthorized use of creative property, and finally Part 6 provides recommendations to address emergent issues.

The two case-studies explained in Part 3 relate to:

1. Nike's unauthorized use of Indigenous cultural expressions.
2. The ongoing Andersen v. Stability AI lawsuit which was brought against three AI companies for copyright infringement.

These cases were chosen to demonstrate how: (1) the risks of unauthorized use of artworks are pervasive notwithstanding your cultural identity, (2) the current national and international copyright protections are ill-equipped to tackle emergent issues in the sociotechnical sphere, and finally (3) to demonstrate how the burdens of unauthorized use of creative works are disproportionately placed on creators or artists themselves.

Part 1: Understanding GenAI art

In early 2024, the Whitney Museum of Art had a special exhibit about the earliest AI program that made art- AARON. Harold Cohen, a British-born artist, had created this software that could create drawings based on commands given to a computer. Cohen had worked on converting his artistic process into a machine-readable code to program computers to make art. AARON was to be a tool of expression for Cohen, instilling perpetuity and independence to his artistic process. Unlike Cohen's model, which was trained on rules and comprised a mechanical arm and a stylus to generate art (Yavuz, 2024), the current model of generative art based on text-based prompts uses the diffusion model to create art. This process is a neural network that mimics the structure of the human brain (Oxford English Dictionary, n.d.). In a diffusion model, the AI learns through deep learning – a method by which it learns patterns from training data (audio, visual, and textual), while continuously fine-tuning its learning to predict an output that is most closely related to a text-based command, a.k.a. prompt.

The current model used by AI image-generators is trained on image datasets, which comprise thousands of images available on the Internet. Some common text-to-image datasets that support the diffusion model are MS-COCO, LAION-5B, FFHQ (Flickr-Faces-HQ), and CelebA-HQ. Stability AI (the company named in the Andersen v. Stability AI lawsuit) has stated that their diffusion model 'Stable Diffusion', used the image dataset developed by LAION - a nonprofit involved in academic research- which has made a dataset of over 5 billion images it has scraped from the Internet, which also includes copyrighted works (Puri, 2025; Saveri Law Firm n.d.). While an artist himself had programmed AARON to make a likeness of his own art, the art made by AI-image generators is more a derivative work of

art based on all the information it has learned from its training data. Instances like these raise important questions of consent, authorship, originality, and creativity.

Part 2.0: Indigenous Peoples: More than a Simple Definition

The term ‘Indigenous Peoples’ lacks a universally agreed-upon definition under international law and policy. The latter was intentionally decided by the drafters of the United Nations Declaration on the Rights of Indigenous Peoples (UNDRIP), adopted in September 2007. The decision reflects the principle that the identification of Indigenous Peoples is the right of the people themselves—a manifestation of their right to self-identification and a fundamental element of the right to self-determination (OHCHR, n.d.). Furthermore, scholars have agreed that the term ‘Indigenous’ broadly refers to the living descendants of pre-invasion inhabitants of lands now dominated by others, and that they share some crucial commonalities such as their relation to the land and environment and the importance of cultural practices and traditional ways of life (Anaya, 2004, p.3).

Indigenous Peoples, at present, comprise non-dominant sectors of society and are determined to preserve, develop, and transmit to future generations their ancestral territories, ethnic identity, knowledge, and heritage as the basis of their continued existence as peoples, in accordance with their cultural patterns, social institutions, and legal system (José, 1972, pp. 11-13). About 476 million Indigenous Peoples are living in over 90 countries (The World Bank, 2016), including ‘developed Western States’ of Canada, the United States, Australia, and Finland (ILO, 2017, p.9). Despite comprising only about 6% of the global population, Indigenous Peoples account for 19% of the world’s extremely poor population (The World Bank, 2016). The latter compounds the issue of sovereignty over knowledge and cultural heritage, as the misappropriation of said cultural resources can be used to further disenfranchise further and impoverish Indigenous Peoples.

Part 2.1: What is Cultural Heritage?

It can be challenging to define and delineate the contours of something as intangible as ‘cultural heritage’, but we can defer to existing scholarship to help break it down. According to the Mexico City Declaration on Cultural Policies from 1982, culture *sensu largo* is:

the whole complex of distinctive spiritual, material, intellectual and emotional features that characterize a society or social group. It includes not only the arts and

letters, but also modes of life, the fundamental rights of the human being, value systems, traditions and beliefs. (p.41)

Similarly, the idea of ‘heritage’ has been rooted in preservation and continuity. According to UNESCO, certain monuments, buildings, works of art, artifacts, landscapes, etc. have “outstanding universal value”. They are declared to belong to “humanity”, with the intent to safeguard them from destruction and to take active steps towards their conservation. Heritage can take tangible forms such as monuments, publications, and sculptures, as well as intangible forms such as oral traditions, community bonds, philosophies, and language (Harvard University, n.d.).

However, for Indigenous Peoples, the idea of cultural heritage sits at the crux of disseminating important cultural and historical knowledge, as well as of recognizing and preserving a way of life. The way of life in such a case includes social, economic, and legal systems that influence and shape Indigenous Peoples. Presently, the Indigenous and Tribal Peoples Convention, 1989 (ILO) is the sole legally binding international treaty concerning Indigenous rights which, in its Article 23, recognizes that “handicrafts, rural and community-based industries, and subsistence economy and traditional activities of the Peoples concerned, such as hunting, fishing, trapping and gathering, shall be recognised as important factors in the maintenance of their cultures.”

The latter helps offer clearer parameters for identifying the cultural heritage of Indigenous Peoples. It is important to note that culture is a fluid concept, prone to change and adaptation over time. Indigenous Peoples are not a homogeneous group; thus, when it comes to the cultural heritage of Indigenous Peoples, they themselves should be able to determine what comprises cultural heritage. Such action would be most conducive to ensuring that they have sovereignty over their knowledge.

Part 2.2: Sovereignty over Knowledge Dissemination

In the 2009 paper, Tsosie discussed how the right to culture of Native Americans should expand to include the right to control cultural expression and “reclaim Native stories”. Article 31 of the UNDRIP takes the idea of sovereignty over the dissemination of knowledge forward by stating that:

Indigenous Peoples have the right to maintain, control, protect and develop their cultural heritage, traditional knowledge and traditional cultural expressions, as well as the

manifestations of their sciences, technologies and cultures, including human and genetic resources, seeds, medicines, knowledge of the properties of fauna and flora, oral traditions, literatures, designs, sports and traditional games and visual and performing arts. They also have the right to maintain, control, protect and develop their intellectual property over such cultural heritage, traditional knowledge, and traditional cultural expressions. (p.22)

Thus, traditional cultural expressions (TCE), as seen in tangible manifestations such as visual art (sculptures, drawings), literature, and dance forms, have been recognized as essential to the protection of the heritage of Indigenous Peoples. Moreover, TCEs are a harmonious manifestation of artistic and symbolic elements of a way of life and have often been known to have religious significance.

The emergent concept of “Data justice” recognizes that the collection, use, and dissemination of data have political, social, and ethical impacts (Dencik et al., 2019). It further advocates for prioritizing the agency of marginalized communities in data governance. This concept extends to providing respective communities with legal jurisdiction over how their data is collected, processed, stored, and used. This idea centers on the data sovereignty of Indigenous Peoples, especially with respect to their TCE. The relationship between documenting cultural heritage for preservation and protection of Indigenous Peoples’ right to control their cultural expressions has, in recent times, been fraught with some tension due to the digitization of records (Prażmowska, 2020, p. 121). TCE has been documented by academics and scholars studying Indigenous Peoples for anthropological research and for archival purposes, through photographs and videos for museum repositories. With the rise in Indigenous tourism, many pictures and videos of various communities are put in the digital sphere. Putting TCE in the digital sphere exposes them to the possibility of being compiled by image-datasets, thereafter, used to train AI software. Even in the absence of AI software, TCE has been borrowed by businesses in the past, leading to non-consensual use that is not only culturally insensitive but also harmful.

Part 2.3: Cultural Appropriation

In a study conducted in 2019, the majority of the Native Americans interviewed indicated negative sentiments towards cultural appropriation (CA), with reasons for opposing it being that it was hurtful, harmful, insensitive, ignorant, causing

objectification and discrimination (Keene et.al., 2023, pp.9-11). It is important to understand what comprises CA and why it is considered harmful. According to WIPO's Glossary of Key Terms, "misuse" is explained as improper use that leads to distorting the meaning or purpose of something. In contrast, the term "misappropriation" is used to indicate the wrongful or dishonest use of another person's property, which may result in the derivation of undue benefits (WIPO, 2012, pp. 27-28). CA combines these two meanings and adds an element of power dynamics to the equation. It has been defined as:

the act by a member of a dominant culture of taking TCE whose holders belong to a minority culture or a group that suffered a history of discrimination, colonization, or exploitation, and repurposing it in a different context, without the authorization, acknowledgement, and/or compensation of the TCE holders' (Venzina, 2019, p.6)

Thus, CA leads to the misrepresentation of TCE and arises from the power imbalance between the taker and the owner of TCE. It occurs without the express consent of communities that own the knowledge.

Part 3: Case Studies

The following case studies focus on two separate but important issues: first, we discuss a case of CA of Indigenous TCE, and second, we discuss emergent issues of IP protection vis-à-vis GenAI technology companies.

Part 3.1: Nike's Unauthorized use of 'Mola' Patterns

In 2019, Nike released a trailer for the Air Force 1 Low "Puerto Rico" model (Figure 3) that used the traditional 'mola' design (Figure 2) of the Guna community without their consent (BBC, 2019). The Guna people are the Indigenous community living in Panama and Colombia. In Panama, they primarily inhabit the San Blas Islands, which are part of the Guna Yala autonomous region (BBC, 2019). According to the Mola Museum (Museo de la Mola) in Panama City, "Molas are rectangular fabric panels that form the chest and back of the blouses of Guna women-wear". These panels are handmade using a reverse appliqué technique. The geometric patterns used in the molas draw inspiration from nature, using motifs of "flowers, birds, reptiles, animals" (Nalewicki, 2022).

Figure 2

A textile depicting the traditional ‘mola’ pattern.



Note. The image was accessed from the website of the Mola Museum (MuMo), Panama City, Panama
(<https://museodelamola.org/en/>)

Figure 3

The Nike Air Force 1 Low “Puerto Rico” model.



Note. The image was accessed from a blog post on ABS-Canada’s website about Nike’s Air Force 1 Low “Puerto Rico” model.
(<https://abs-canada.org/nike-shoe-design-stolen-from-guna-indigenous-women-and-marketed-as-puerto-rican/>)

Following grassroots mobilization by the Guna women, Panama introduced Law 20 in 2000 to protect TCE and traditional knowledge. This law is “designed to protect traditional dress, music, dance and major indigenous [sic] handicrafts such as tagua nut carvings, hand-beaded chaquira necklaces and chacara woven bags” (WIPO Magazine, 2005). Panama is among the few countries in the world that recognize the rights of Indigenous Peoples over their IP. These IP protections are not blanket and require formal registration. In 2002, the Guna people registered their “collective right” over the mola, which bestows upon them the sole right to commercial use and application of the garment (Obaldia, 2005).

After Nike announced the shoe, the Guna people protested the unauthorized use of their TCE through their lawyer, Aresio Valiente. Valiente is a Guna lawyer and one of the drafters of Law No. 20 (Parker, 2019). Not only did Nike fail to credit the Guna people for the artwork, but it also misrepresented the design as inspired by Puerto Rican art. The group sought the withdrawal of the shoe and financial compensation for the damage to their cultural identity (BBC, 2019). While Nike withdrew the shoe and apologized for the inaccurate representation, no information about any financial compensation is available.

This case demonstrates that it is possible to pinpoint accountability when a manufacturer, fashion house, or other creator plagiarizes a copyrighted work. Who is to be held accountable when a creative work is made using GenAI software that is trained on millions of images found online? Some important arguments and counterarguments have emerged from the ongoing class-action lawsuit, *Andersen v. Stability AI*.

Part 3.2: Andersen v. Stability AI

In December 2022, cartoonist and illustrator Sarah Andersen wrote a guest essay for the New York Times about the appropriation of her art. She brought to light the issue of AI-generated art mimicking her artistic style (Figures 4 & 5) (Andersen, 2022). Earlier that year, artist and illustrator Karla Ortiz had deep-dived into how text-to-image GenAI software works and was shocked to find that her artwork had been used to train datasets for AI models (Salkowitz, 2024). In August of that year, Kelly McKernan, another visual artist, found out that her name was being used as a text prompt to feed into text-to-image AI generators to create a likeness of her work (Chayka, 2023). All three creators had learned that their work was being used without consent, compensation, or credit (Butterick, as cited in Chayka, 2023).

Figure 4

An original Sarah Andersen comic, demonstrating her artistic style.



Note. The above comic appeared in a Guest Essay that Sarah Andersen wrote for the New York Times in December 2022

(<https://www.nytimes.com/2022/12/31/opinion/sarah-andersen-how-Algorithm-took-my-work.html>)

Figure 5

An AI-generated comic mimicking Andersen's artistic style.



Note. The above comic appeared in a Guest Essay that Sarah Andersen wrote for the New York Times in December 2022

(<https://www.nytimes.com/2022/12/31/opinion/sarah-andersen-how-Algorithm-took-my-work.html>)

Soon after, in January 2023, the illustrators, along with other artists, sued three AI companies- Stability AI, Inc., Midjourney, Inc., and DeviantArt, Inc.- for copyright infringement. The class-action lawsuit is now called Andersen v. Stability AI. The lawsuit alleges that AI-image generation products were trained on the LAION-5b database, which consists of 5 billion images scraped from the Internet without express permission from creators (Saveri Law Firm n.d.). This publicly available dataset is developed by LAION – a nonprofit entity (Chayka, 2023). On August 12, 2025, a U.S. District Judge found that claims of both direct and indirect copyright infringement were plausible, and the case has since moved to the discovery stage (Whiddington, 2024).

It is interesting to note that not all the images in the database would be copyrighted; some could also be under other licenses like Creative Commons licenses, commercial licenses, Terms of Use, etc. (Downing, 2023). This nuance would further muddy the waters of placing accountability. Downing (2023) argues that an AI-image generator “interpolates ALL the images in the training data to create its output.” What this means is that the image finally generated cannot be traced back to one source image, but instead is derived from all images. The latter is also understood as the “black box problem” of deep-learning systems wherein we are unable to reverse engineer why a system made a certain decision (Rawashdeh as cited in Bloin, 2023, para.3). What is clear, however, is that this ongoing lawsuit would set the precedent in dealing with questions of fair use, consent, and rights of creators vis-à-vis technology companies.

As the lawsuit grapples with whether copyright protections are sufficient to safeguard artists, safeguarding artworks in the absence of formal copyright protections has become vital as well. Works by upcoming artists, as well as TCE (art, music, sculpture, etc.) of Indigenous Peoples, could make their way into AI training datasets without their express consent or knowledge. The issue of non-consensual usage becomes especially pertinent when there is a probability of CA of art made by Indigenous People.

Part 4: What Protections Exist and Why They Fail?

A simple text prompt to an AI software to transform a picture into another artist’s distinctive style can create near-perfect derivative works. This, despite the fact that some of the original artwork is protected by copyright law, helps demonstrate that current AI software cannot properly discern what constitutes protected works. This increases the risk of CA or even non-volitional misuse of TCE available on the Internet by users of GenAI software. Users could be oblivious to the presence of copyright protections on certain artworks or be uninformed about the symbolic uses of certain motifs used by Indigenous Peoples. Such users could accidentally deploy text prompts that could produce egregious or even offensive results. Since 2025, the Australian government has formed a First Nations Expert Working Group after Aboriginal artists pointed out how some AI software inaccurately represented their work (Vyver & Jash, 2025). In addition, it is also necessary to take a deeper look into the power asymmetries in the process of negotiating consent for the circulation

of traditional knowledge (Anderson, Duke University, 2010, p.25) and how these imbalances can lead to CA.

The present system of IP protections for certain TCE, more specifically handicrafts made by Indigenous Peoples, comes from the Paris Convention for the Protection of Industrial Property, the Berne Convention for the Protection of Literary and Artistic Works, and the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) (WIPO, 2002). Handicrafts are categorized as ‘works of applied art’ and fall under the purview of objects eligible for protection as intellectual property. This allows handicrafts to be granted protections through the following:

- a. Geographical Indications/ Trademarks for location or community-specific handicrafts and produce.
- b. Copyright for protecting visual or textual traditions.
- c. Patents for know-how or traditional knowledge
- d. However, each of these tools has its own shortcomings and is not fully able to realize the goal of protecting TCE.

Geographical indications provide protection only within a territory, but appropriation can be committed by entities outside territorial boundaries; e.g., Nike is based in the United States of America, but the Guna People are spread across several States in the Caribbean. Moreover, copyright law provides protections based on ‘originality’, attributed to a single owner. Most TCEs are collectively owned, passed down through customary practices or oral tradition, and are often ineligible for copyright protection (Muthukumar & Ravenscroft, 2025). While industrial design and patent laws can protect against outright copying of handicrafts, it would be difficult to clearly identify plagiarism when elements or motifs are used as inspiration. The atmosphere is certainly ripe for discussions on how the burden of proving CA by an AI software can be reduced.

Part 5: Of Burdens and Benefits

The GenAI image generator market, valued at USD 419 million in 2024, is expected to reach USD 1,880 million by 2033, with a 18.1% growth rate (Dimension Market Research, 2024). Stability AI is a significant player in this market. According to Valyaeva (2023), between 2022 and 2023, 15 billion images were created using GenAI software, of which 80% were created using Stability AI’s model. The author notes that the number “15 billion” is significant because “it took photographers a

few dozen years, from the first photograph taken in 1826, to reach that mark”. After OpenAI launched DALL-E, an average of 34 million images is being created by users daily (Valyaeva, 2023). The latter demonstrates the extent of power and influence that technology companies have in the GenAI image market.

It is important to note that these platforms operate on a freemium model where users can access basic features but must pay for premium features. These are revenue-generating sources for technology companies; however, this revenue does not flow as compensation to actual creators. As these platforms become more accessible, there are bound to be massive increases in the number of images generated – leading to higher chances of derivative works, from known and unknown artists, flooding digital spaces.

In 2022, Gabriel argued that since AI has deeply integrated into our social, political, and economic systems, our current world can be called a sociotechnical system (pg. 219). As a sociotechnical system, AI influences our interactions with the State and with each other; such a system is well-suited for analysis under the Rawlsian lens of justice. He writes, “what matters for individual liberty is that practices that have the potential to interfere with this freedom are appropriately regulated so that infractions do not come about” (p. 228).

According to Rawls (1971), in a truly just society, the burdens and benefits would be so distributed that the least advantaged would be made better off (p. 13). When AI models use Indigenous knowledge to train and fine-tune themselves, and when that knowledge is synthesized into something altogether new, it becomes important to understand how the benefits or burdens of this dissemination of knowledge impact the communities that own it. Michael Running Wolf, who runs a Lakota AI camp, expresses concern over the fact that AI is completely able to bypass humans in the creation of art, especially when it comes to Indigenous Peoples, as there are images of Indigenous artisanal work online that can be used by AI models to create derivative works (Hendrix, 2023). The fears of CA are not unfounded.

Presently, we do not have a global policy that clearly identifies how Indigenous knowledge can be protected. The risk of CA is high when technology companies can use information on the Internet while dodging issues of IP protection and consent. This exposes Indigenous knowledge to misinterpretation and misrepresentation, thereby adding an unnecessary burden on said communities to (1) fight for sovereignty over knowledge management, and (2) do so in the absence of globally

recognized legal protections. The latter is not only inherently unfair, but also places the burden of protection of traditional knowledge squarely on Indigenous communities instead of setting a stringent requirement on technology companies to prioritize consent in the generation of datasets. By making a marginalized group worse off, we can see how the balance in this sociotechnical system is skewed in favor of a more advantaged group – technology companies. This fortifies the case for a new theory of justice for a digital world.

Part 6: Recommendations

The central theme of this paper was to identify how the emergence of GenAI creates an unequal sharing of burdens and benefits, specifically with regard to TCE of Indigenous Peoples. There is certainly a need for a proactive approach to anticipate cases of injustice, in the form of CA or misuse, and prevent their occurrence *ex-ante*. Some plausible pathways that can achieve this end are:

- A spirit of ‘different but equal’ should be imbued in IP protections so that distinct facets of the cultural heritage of Indigenous Peoples are recognized and protected. This can be achieved by reimagining copyright through the lens of collective ownership, customary law, and perpetual protection (Muthukumar & Ravenscroft, 2025). Moreover, given the transnational character of Indigenous Peoples and the globalized nature of business, there is space to develop and codify an IP law that is international in jurisdiction. An alternate approach could be creating Indigenous knowledge-specific licenses based on principles like the Creative Commons license (Anderson, 2010). This would accommodate the need for protecting Indigenous knowledge, decentralize knowledge management, and expand the parameters of fair compensation.
- Building ethical datasets by using the humans-in-the-loop approach. An AI does not have a moral code, and biases in the dataset will translate into the AI’s learnings. A humans-in-the-loop approach propagates “selective inclusion of human participation” (Wang, 2019, October 21) in process automation. Thus, ensuring diversity in programmer groups and regular audits of the AI algorithm can help de-bias the data and ultimately the AI software (Henz, 2021). Including Indigenous Peoples in the database creation process not only ensures de-biased data-labeling but also acknowledges their data sovereignty.

- Bottom-up AI initiatives can guarantee sovereignty over a knowledge system, its dissemination, circulation, and use. When discussing bottom-up AI initiatives, the work of Te Hiku Media, an Indigenous media hub based in New Zealand, is notable for creating its own large language model (LLM) (Moreno, 2024). Significant strides in data sovereignty have also been made by the Indigenous AI Lab, being built by Suzanne Kite, an Oglála Lakhóta artist, composer, and researcher (Durón, 2024). Such endeavors ensure that decisions about the technological, economic, and political use of AI are grounded within Indigenous values. This case is proof that AI can be used as a force for good to broaden the benefits of digital technologies.
- Moral suasion could be used as a tool to shape AI use. In 2024, The Authors Guild released guidelines for AI use for members. One of the guidelines was:
Respect the rights of other writers when using generative AI technologies, including copyrights, trademarks, and other rights, and do not use generative AI to copy or mimic the unique styles, voices, or other distinctive attributes of other writers' works in ways that harm the works.

Similarly, other creative associations can issue moral appeals to build a system of self-regulation among members of industry.

Conclusion

The emergence of AI has altered the way creative content is produced and consumed. Pending clear jurisprudence on the use of copyrighted material in the digital sphere for training AI models, it is necessary to identify the risks posed by unbridled AI to traditional knowledge and TCE. In doing so, we would have to grapple with questions about the sharing of burdens and benefits of AI technology. Should the burden of ethical use fall on the user alone? How should we navigate instances where cultural appropriation occurs from TCE belonging to multiple Indigenous Peoples? In such cases, the procedures for adequate amends and fair reimbursement would have to be settled through discussion with impacted communities.

Furthermore, can we explore a solution where AI software itself is used to detect unauthorized use, misuse, and misrepresentation? Solutions can be found that are rooted in the spirit of 'collective ownership', data sovereignty, and inclusivity. The march of AI cannot be stalled; our systems would have to adapt. Going forward, it

is inevitable that new paradigms of justice and fairness will have to be infused into the technological, legal, and political systems to ensure that everyone can benefit.

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